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Homework 6: Phase Retrieval

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The assignment report is due on 25 November, 2026. In it, you must solve the exercises presented in the guide in report format, and attach the files you generate, such as .py, ipynb, .mat, .m, .fig, etc. You also have to include a video connecting your report and the developed code. The report should briefly describe the experiments performed, including the following sections: summary, description of experiments, and conclusions. It is suggested that you prepare the report in LaTeX to facilitate writing equations. Send a compressed folder with the files to the Aula submission corresponding unit: , with the subject line: **Homework6_FotografíaComputacional** and the compressed folder's name **homework6_nameSurname.zip**, where Name is your first name and Surname is your last name. Caution: any attempt at copying or fraud will invalidate the lab and result in a zero on the lab grade.

1. Learning Goals

To simulate a coded illumination system to recover the amplitude and phase of an object using complementary coded apertures.

2. Introduction

In computational photography, phase retrieval is the problem of reconstruction of a signal or object from the magnitude of its Fourier transform, where the phase information is missing. Phase retrieval emerges in various fields of science and engineering, including electron microscopy, crystallography, astronomy, and optical imaging. To understand the importance of the phase, see Figure. 1, where two pair of images are depicted cameraman and lena, if the Fourier transform of the two imaged is computed and the phases are interchanged between the two images the recovered images using the inverse Fourier transform preserves the details of the image in the phase despite its magnitude.

2.1. Forward Model

Figure 2 illustrates the coded illumination process. A collimated coherent beam illuminates an object, producing an unknown complex wavefront **O**. This wavefront is encoded by a set of coded apertures, denoted

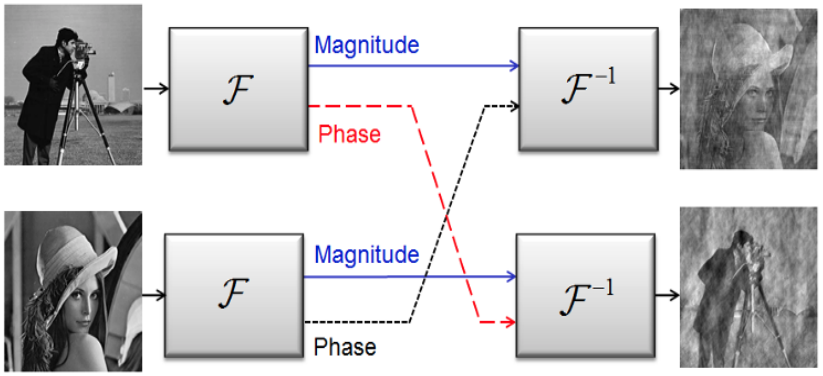


Figura 1: If the Fourier transform of two different images are interchange, then computing the Fourier inverse the information phase is preserved.

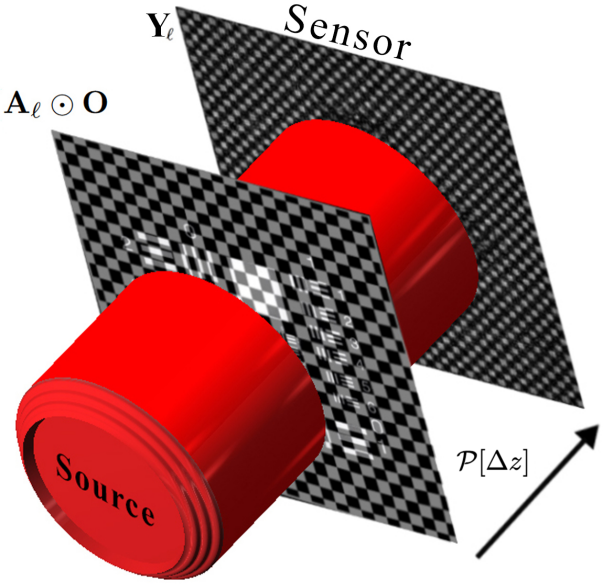


Figura 2: Schematic representation of the coded illumination system for compressive phase retrieval. A coherent wavefront from the complex object $\mathbf{O}$ is modulated by the ℓth coded aperture $\mathbf{A}_\ell$, propagates a distance z in free space, and produces the intensity measurement $\mathbf{Y}_\ell$ according to Eq. (3). This measurement contains phaseless information that is used in the phase retrieval algorithm.

as $\mathbf{A}_\ell$, where $\ell = 1, 2, \ldots, L$, representing a total of L measurements. The encoding process is represented as $\mathbf{A}_\ell \odot \mathbf{O}$, where $\odot$ denotes the Hadamard product. After free-space propagation over a distance Δz along the z axis, a diffraction pattern $\mathbf{I}_\ell$ reaches an image sensor. This can be mathematically expressed as

$$\mathbf{I}_\ell = \mathcal{P}[\Delta z]\left\{\mathbf{A}_\ell \odot \mathbf{O}\right\}, \tag{1}$$

where $\mathbf{I}_\ell, \mathbf{O} \in C^{d\times d}$ and $\mathbf{A}_\ell \in \{0,1\}^{d\times d}$, with $d \times d$ denoting the number of pixels in the CA. The operator $\mathcal{P}[\Delta z]$ represents the near-field propagator, by the convolution

$$\mathbf{U}(u,v) = \left[\frac{e^{ik\Delta z}}{i\lambda\Delta z} e^{i\frac{k}{2\Delta z}(u^2+v^2)}\right] \otimes \mathbf{U}_0, \tag{2}$$

where $\mathbf{U}(u,v)$ is the field at the observation plane with coordinates (u,v), and $\mathbf{U}_0$ is the field at the source plane with coordinates (x,y). Here, λ denotes the wavelength of the source, and $k = 2\pi/\lambda$ is the wavenumber.

Therefore, the ℓth coded measurement of the diffraction pattern, recorded at the image sensor, is proportional to the absolute square of the complex wavefront,

$$\mathbf{Y}_\ell = |\mathbf{I}_\ell|^2 + \mathbf{\Omega} \quad (\mathbf{\Omega} \in R^{d\times d}), \tag{3}$$

where $|\cdot|$ represents the element-wise modulus operator, and $\mathbf{\Omega}$ is an additive noise term in the real domain $R^{d\times d}$. Before explaining the nature of the measurement, we can describe the complex object reconstruction as the optimization problem:

$$f(\mathbf{O}) = ||\mathbf{Y}_\ell - \mathbf{A}_\ell(\mathbf{O}\mathbf{O}^H)||_2^2, \tag{4}$$

where $\mathbf{Y}_\ell$ is the phaseless data, $\mathbf{A}_\ell$ is the coded aperture and $\mathbf{O}$ is the complex signal to be recovered. To obtain a well-conditioned inversion problem, we need to find a subset L of coded apertures $\mathbf{A}$ that minimizes the correlation between the measurements $\mathbf{Y}_\ell$. Hadamard matrices are well known for generating maximal sets of pairwise independent random variables with uniform measure.

3. Exercises

1. To start, check the following Github repository; the task is to implement the MATLAB code in Python denominated phase-retrieval.
2. For the simulations, use only random complementary masks with 4 and 16.
3. Simulate using the hybrid input-output algorithm (HIO).
4. For homework 6, use the greyscale dataset and select 24 images; use all images as amplitude and phase. For this task, use only grayscale images from the greyscale dataset.
5. To evaluate spatial image quality using Peak-Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM).
6. To plot the spatial quality metrics for the selected 24 grayscale images.

   The Peak Signal-to-Noise Ratio (PSNR) in decibels (dB) is given by:

   $$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}_I^2}{\text{MSE}}\right), \tag{5}$$

   where MAX_I represents the maximum possible value for the image pixel (e.g., 255 for an 8-bit image). The mean squared error is given by

   $$\text{MSE} = \frac{1}{m \cdot n}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}[I(i,j) - K(i,j)]^2, \tag{6}$$

   $$\text{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \tag{7}$$

   where:

   - μ_x and μ_y are the local means (luminance averages) of x and y.
   - σ_x^2 and σ_y^2 are the local variances (contrast) of x and y.
   - σ_{xy} is the local covariance (structural similarity) of x and y.
   - $C_1 = (K_1 L)^2$ y $C_2 = (K_2 L)^2$ are stabilizing constants.
   - L is the dynamic range of the pixel values (typically $2^B - 1$, where B is the number of bits per pixel).
   - $K_1 \ll 1$ y $K_2 \ll 1$ are small scalar contants.

   PSNR SSIM